

Design & Analysis of Algorithms

Lab 1: Algorithm Performance Analysis Workbench

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Q1. 1.1 Recursive and Iterative Computation Analyzer (10 marks)

```
def factorial_recursive(n):
    if n == 0 or n == 1:
        return 1
    else:
        return n*factorial_recursive(n-1)
def factorial_iterative(n):
    fact = 1
    for i in range(1, n+1):
        fact*=i
    return fact

def fibonacci_iterative(n):
    a,b = 0,1
    for _ in range (1, n+1):
        a,b = b,a+b
    return a
def fibonacci_recursive(n):
    if n == 0:
        return 0,1
    if n == 1:
        return 1,1
    f1,c1 = fibonacci_recursive(n-1)
    f2,c2 = fibonacci_recursive(n-2)

    return f1 + f2, c1 + c2 + 1

def analyze_recursive_iterative(n):
    fact_recursive = factorial_recursive(n)
    fact_iterative = factorial_iterative(n)
    fib_recursive, fib_count = fibonacci_recursive(n)
    fib_iterative = fibonacci_iterative(n)

    result = []

    result.append("Computation Analysis Report")
    result.append("Recursive Factorial: " + str(fact_recursive))
    result.append("Iterative Factorial: " + str(fact_iterative))
    result.append("Recursive Fibonacci: " + str(fib_recursive))
    result.append("Iterative Fibonacci: " + str(fib_iterative))

    result.append("Operation Count Comparison")
    result.append("Recursive Factorial Count: " + str(n+1))
    result.append("Iterative Factorial Count: " + str(n))
    result.append("Recursive Fibonacci Count: " + str(fib_count))
    result.append("Iterative Fibonacci Count: " + str(n))

    return result
```